

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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NAME : Karan Hanumant Katare

COURSE: BFSI(Online)

SEMESTER : 3rd

PERSONAL FINANCE ADVISOR BOT: AI-POWERED PERSONAL FINANCE MANAGEMENT PLATFORM

Project Description :-

Personal Finance Advisor Bot is an AI-powered personal finance management and advisory platform designed to help users manage their personal income, expenses, budgets and savings goals in one place.

The platform provides a personalized financial workspace where users can securely create an account, record their income and expenses, organize transactions into categories, create monthly budgets, monitor savings goals and generate monthly financial reports.

The system also provides an intelligent AI Financial Advisor that uses the user's own financial records to provide educational financial guidance and recommendations. The AI service supports Google Gemini and OpenAI as AI providers, while a deterministic rule-based fallback provides recommendations when an external AI API is unavailable or not configured.

The application provides an interactive dashboard containing financial summary cards, charts and budget status indicators. Users can monitor their financial condition through visual representations of income, expenses, category-wise spending, budgets and savings.

The system also maintains AI recommendation history and allows users to access an advisor chat, generate monthly reports and export reports in formats such as print/PDF/CSV.

The project is developed as a production-oriented web application using Python, Flask, Jinja2, SQLAlchemy, Flask-Login, Flask-WTF, Bootstrap 5 and Chart.js, with SQLite for local development and support for PostgreSQL-style deployments.

Technology & AI :-

Frontend:

- **HTML5**
- **CSS3**
- **Jinja2 Templates**
- **Bootstrap 5**
- **JavaScript**
- **Chart.js**
- **Responsive FinTech User Interface**

Backend:

- **Python**
- **Flask**
- **Flask Application Factory**
- **Flask Blueprints**

Database:

- **SQLite for local development**
- **SQLAlchemy ORM**
- **PostgreSQL-compatible database support for production**

Authentication:

Flask-Login

- **Werkzeug Password Hashing**
- **Secure User Sessions**
- **Protected Routes**

Security:

- **Flask-WTF CSRF Protection**
- **Password Hashing**
- **Secure Session Management**
- **Environment Variables for API Keys**
- **User Ownership Isolation**

AI:

- **Google Gemini API**
- **OpenAI API**
- **Rule-Based AI Fallback**
- **AI Recommendation History**

Visualization:

- **Chart.js**
- **Dashboard Financial Charts**
- **Budget Status Indicators**

Testing:

- **Pytest**
- **Automated Application Tests**

Deployment:

- **Application Container / Web Application Deployment**
- **Health Check Endpoint**

Version Control:

- **Git**
- **GitHub**

The GitHub project specifically documents Flask/Jinja2, SQLAlchemy, Flask-Login, Flask-WTF, Bootstrap 5 and Chart.js, with Gemini/OpenAI support and a local rule-based fallback.

Key Features :-

- **Secure user registration and login system**
- **Personal finance dashboard**
- **Income management and tracking**
- **Expense management and tracking**
- **Category-based financial organization**
- **Monthly budget planning**
- **Savings goal management**
- **Savings contribution history**
- **AI-powered financial advisor**
- **AI advisor chat**
- **Personalized financial recommendations based on user's own records**
- **Google Gemini AI integration**
- **OpenAI integration**
- **Rule-based AI fallback when external AI services are unavailable**

Monthly financial reports

- **Print, PDF and CSV report output**
- **Interactive financial charts using Chart.js**
- **Budget status indicators**
- **AI recommendation history**
- **Profile management**
- **Shared month selector across financial modules**
- **Multiple financial personas including Salaried, Student, Freelancer and Household**
- **Protected authenticated routes**
- **User ownership isolation**
- **Database persistence**
- **Responsive fintech-style interface**
- **Health check endpoint for deployment**
- **Demo data and seed functionality**

The repository's documented routes include `/dashboard`, `/income`, `/expenses`, `/categories`, `/budget`, `/savings`, `/advisor`, `/reports` and `/profile` for authenticated users.

Financial Management Modules :-

1. Income Management

The Income Management module allows users to record and monitor their different sources of income.

Users can maintain records such as:

- **Salary**
- **Freelance income**

Other income sources

- **Income amount**
- **Income category**
- **Date and description**

The stored income data is used by the dashboard and AI advisor to understand the user's overall financial condition.

2. Expense Management

The Expense Management module allows users to record their daily expenses.

Expenses can be organized into different categories so that users can understand where their money is being spent.

The system can help users identify:

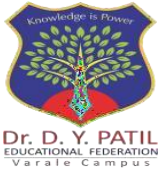
- **Total expenses**
- **Category-wise spending**
- **Monthly spending patterns**
- **High-expense categories**
- **Budget utilization**

This information is also used by the AI advisor to provide personalized financial guidance.

3. Budget Management

The Budget module allows users to define spending limits for different categories.

The system monitors the user's actual expenses against the defined budget and provides budget status indicators.



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This helps users understand whether their spending is within their planned limits.

4. Savings Goals

The Savings module allows users to create financial savings goals and track their progress.

Users can define a target amount and record contributions toward that target.

The system maintains contribution history and helps users monitor progress toward their savings objectives.

5. AI Financial Advisor

The AI Financial Advisor is one of the main features of the project.

It analyzes the user's financial records and provides educational financial guidance based on available information.

The AI can help users understand:

- **Spending patterns**
- **Budget management**
- **Savings opportunities**
- **Financial habits**
- **Monthly financial condition**
- **Areas where expenses can be controlled**

The application supports Gemini and OpenAI providers and uses a deterministic rule-based fallback if the external AI provider is unavailable or no API key is configured.

Important: The project's AI guidance is educational and is not intended to be regulated financial, tax or investment advice.

6. Financial Dashboard

The dashboard provides a centralized view of the user's financial information.

It displays important financial information such as:

- **Income summary**
- **Expense summary**
- **Budget status**
- **Savings information**
- **Financial charts**
- **Monthly financial information**
- **Category-based spending information**

Chart.js is used to create interactive financial visualizations.

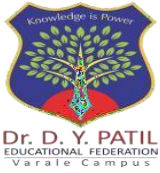
7. Monthly Reports

The Monthly Reports module allows users to generate a summary of their financial activity.

The report can contain information about:

- **Monthly income**
- **Monthly expenses**
- **Category-wise expenses**
- **Budget information**
- **Savings**
- **Financial recommendations**

Reports can be printed or exported in PDF/CSV-compatible formats.



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Scenarios :-

Scenario 1: Student Managing Monthly Expenses

A student registers on the Personal Finance Advisor Bot and enters their monthly income and expenses.

For example, the student records their income from allowance or part-time work and enters expenses such as food, travel, education and entertainment.

The dashboard summarizes the financial information and displays category-wise spending through charts.

The AI advisor can then provide educational suggestions for improving budgeting and controlling unnecessary expenses.

Scenario 2: Salaried User Planning a Monthly Budget

A salaried user enters their monthly salary and regular expenses.

The user creates budgets for categories such as:

- **Food**
- **Transportation**
- **Shopping**
- **Bills**
- **Entertainment**

The system compares spending with the planned budget and displays budget status indicators.

The AI advisor uses the user's financial records to provide personalized educational recommendations for better money management.

Scenario 3: User Creating a Savings Goal

A user wants to save money for a specific financial goal.

The user creates a savings goal and enters the target amount.

The user then records contributions over time.

The Savings module maintains contribution history and allows the user to monitor progress toward the target.

The AI advisor can provide general suggestions related to savings and budgeting based on the user's recorded financial information.

Scenario 4: User Analyzing Monthly Financial Performance

A user returns to the application at the end of the month.

The user opens the Reports section and generates a monthly financial report.

The system provides a summary of income, expenses and financial activity.

The user can view charts and export the report for future reference.

The AI advisor can also provide educational recommendations based on the recorded financial information.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objectives of the Personal Finance Advisor Bot, identify target users, understand personal finance management requirements and finalize the major modules such as income, expenses, budgets, savings, reports and AI advisory.

Milestone 2 – Environment Setup & Backend Development

Set up the Python development environment and Flask application.

Implement the Flask application factory and Blueprint-based application structure.

Configure SQLAlchemy for database operations and create the required financial data models.

The project provides local setup using a Python virtual environment, installation of requirements and execution through run.py.

Milestone 3 – Authentication & Security Layer

Develop user registration and login functionality.

Implement:

- **User authentication**
- **Password hashing**
- **Protected routes**
- **Session management**
- **CSRF protection**
- **User ownership isolation**
- **Secure configuration using environment variables**

Flask-Login, Werkzeug password hashing and Flask-WTF CSRF protection are documented components of the project.

Milestone 4 – Finance Management Modules

Implement the core financial modules:

- **Income Management**
- **Expense Management**
- **Categories**
- **Budget Management**
- **Savings Goals**

Contribution History

- **Financial Dashboard**

Develop CRUD operations so users can create, view, update and manage their personal financial records.

Milestone 5 – AI Advisor, Dashboard & Reports

Integrate the AI financial advisor with supported AI providers.

Implement:

- **Gemini integration**
- **OpenAI integration**
- **Rule-based fallback**
- **Advisor chat**
- **AI recommendation history**
- **Dashboard charts**
- **Monthly reports**
- **Print/PDF/CSV output**

The project supports automatic provider selection through configuration and keeps AI API keys server-side rather than exposing them to frontend code.

Milestone 6 – Testing, Optimization & Deployment

Perform automated testing using Pytest.

Test important application areas including:

- **Registration and login**
- **Protected routes**

Income CRUD

- **Expense CRUD**
- **User ownership**
- **AI fallback**
- **AI budget generation**
- **Reports**
- **Dashboard chart data**
- **Database persistence**
- **Different user personas**

The repository documents a Pytest suite covering these areas.

Finally, optimize the application and deploy it as a web application with a health-check endpoint.

System Workflow :-

User Registration/Login



Personal Finance Dashboard



Add Income & Expenses

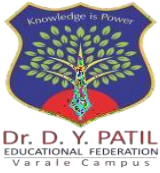


Organize Transactions into Categories



Create Monthly Budgets





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Create & Track Savings Goals



System Stores Financial Records



Dashboard Generates Financial Visualizations



AI Advisor Analyzes User Financial Records



Personalized Educational Financial Guidance



Generate Monthly Reports



Export / Print Financial Reports

AI Working :-

The AI component works by using the user's financial information as context for generating financial guidance.

Step 1:

The user enters income, expenses, budgets and savings information.

Step 2:

The application stores the financial information in the database.

Step 3:

When the user requests financial guidance, the AI service collects relevant financial context from the user's records.

Step 4:

The configured AI provider processes the request.

The project supports:

Google Gemini → OpenAI → Rule-Based Fallback

depending on the configured provider and availability.

Step 5:

The generated recommendation is displayed to the user through the Advisor interface.

Step 6:

AI recommendations can be stored in the recommendation history for future reference.

Database & Data Management :-

The project uses SQLAlchemy ORM for database management.

The application supports SQLite for local development and PostgreSQL-compatible database deployment for production environments.

The system stores information related to:

- **Users**
- **Income**
- **Expenses**
- **Categories**
- **Budgets**
- **Savings Goals**

- **Savings Contributions**
- **AI Recommendations**

The project also includes database persistence and ownership-isolation testing to ensure users access their own financial records.

Security Features :-

The Personal Finance Advisor Bot includes multiple security mechanisms:

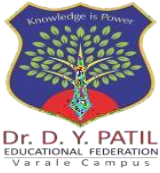
- **Password hashing using Werkzeug**
- **User authentication using Flask-Login**
- **Protected authenticated routes**
- **Flask-WTF CSRF protection**
- **Secure session management**
- **Environment variables for sensitive API keys**
- **User ownership isolation**
- **Server-side storage of AI provider credentials**
- **Secure production configuration**

The project documentation specifically recommends using a strong `SECRET_KEY`, durable production database configuration, secure cookies and keeping `.env`, databases and provider credentials out of source control.

Testing :-

Automated testing is implemented using Pytest.

The test suite covers:



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- **User registration**
- **User login and logout**
- **Protected routes**
- **Income CRUD operations**
- **Expense CRUD operations**
- **Ownership isolation**
- **Rule-based AI fallback**
- **AI budget generation**
- **Monthly reports**
- **Dashboard chart payloads**
- **Database persistence**
- **Salaried persona**
- **Student persona**
- **Freelancer persona**
- **Household persona**

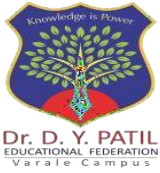
The repository provides the command:

pytest -q

for running the automated test suite.

Expected Outcome :-

The Personal Finance Advisor Bot aims to provide users with a simple and intelligent platform for managing their personal finances.



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The system helps users track income and expenses, plan budgets, monitor savings goals, analyze their monthly financial activity and receive AI-powered educational guidance.

The project demonstrates practical knowledge of:

- **Python**
- **Flask**
- **Jinja2**
- **SQLAlchemy**
- **Database Management**
- **User Authentication**
- **Web Development**
- **REST-style Application Architecture**
- **AI API Integration**
- **Google Gemini**
- **OpenAI**
- **Rule-Based AI**
- **Chart.js**
- **Financial Data Visualization**
- **Automated Testing**
- **Secure Application Development**
- **Git and GitHub**
- **Cloud/Web Deployment**

The project also provides a foundation for future improvements such as advanced financial analytics,

additional AI providers, improved financial forecasting, mobile application integration and connection with external financial data sources.

Conclusion :-

The Personal Finance Advisor Bot provides an integrated solution for personal finance management by combining traditional financial tracking features with Artificial Intelligence.

The application allows users to record income and expenses, manage categories, create budgets, track savings goals and generate monthly reports through a single platform.

The integration of an AI Financial Advisor provides users with personalized educational guidance based on their own financial records. The system also includes a rule-based fallback mechanism so that basic financial recommendations can still be provided when external AI services are unavailable.

With its Flask-based backend, SQLAlchemy database layer, secure authentication, responsive interface, Chart.js visualizations and AI integration, the project demonstrates how modern web technologies and Artificial Intelligence can be combined to build a practical personal finance management platform.

The project successfully demonstrates the practical application of AI, Python, web development, database management, authentication, data visualization and automated testing in the FinTech domain.